

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n AS maxCommitSha\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

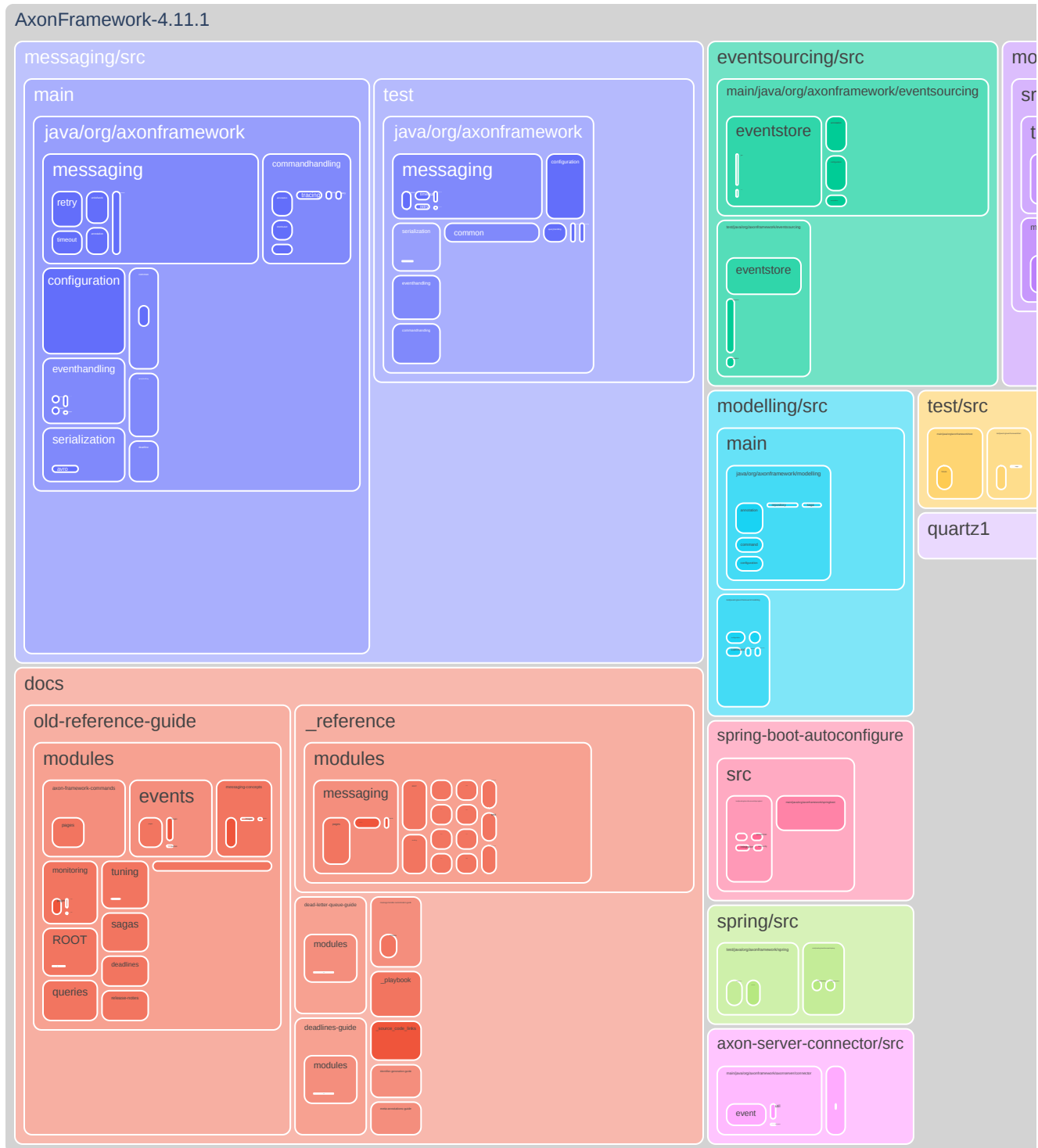
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	305.000000	305.000000	305.000000	305.000000	305.000000	305.000000
mean	14.062295	7.636066	348.750820	412.901639	651.790164	583.085246
std	57.455641	7.093502	1465.259193	1048.994357	1222.531267	1231.770321
min	1.000000	2.000000	2.000000	0.000000	0.000000	0.000000
25%	1.000000	3.000000	24.000000	18.000000	55.000000	17.000000
50%	4.000000	6.000000	75.000000	42.000000	262.000000	104.000000
75%	8.000000	8.000000	187.000000	250.000000	347.000000	292.000000
max	904.000000	71.000000	22225.000000	4791.000000	4879.000000	4790.000000

	directoryPath	directoryParentPath
0	AxonFramework-4.11.1/.idea/copyright	AxonFramework-4.11.1/.idea
1	AxonFramework-4.11.1/.idea/inspectionProfiles	AxonFramework-4.11.1/.idea
2	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src/test/java/org/axonframework/messaging	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src test/java/org/
3	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/command	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector
4	AxonFramework-4.11.1/core/src/test/java/org/axonframework/commandhandling/disruptor	AxonFramework-4.11.1/core/src/test/java/org/axonframework
5	AxonFramework-4.11.1/core/src/test/java/org/axonframework/contextsupport/spring	AxonFramework-4.11.1/core/src/test/java/org/axonframework
6	AxonFramework-4.11.1/core/src/test/java/org/axonframework/eventhandling/scheduling/quartz	AxonFramework-4.11.1/core/src/test/java/org/axonframework eventl
7	AxonFramework-4.11.1/docs/_playbook/localLinks	AxonFramework-4.11.1/docs/_playbook
8	AxonFramework-4.11.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.11.1/docs/_reference/modules/ROOT
9	AxonFramework-4.11.1/docs/_reference/modules/ROOT/partials	AxonFramework-4.11.1/docs/_reference/modules/ROOT
10	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
11	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
12	AxonFramework-4.11.1/docs/_reference/modules/configuration/pages	AxonFramework-4.11.1/docs/_reference/modules/configuration
13	AxonFramework-4.11.1/docs/_reference/modules/configuration/partials	AxonFramework-4.11.1/docs/_reference/modules/configuration
14	AxonFramework-4.11.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.11.1/docs/_reference/modules/disruptor
15	AxonFramework-4.11.1/docs/_reference/modules/disruptor/partials	AxonFramework-4.11.1/docs/_reference/modules/disruptor
16	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
17	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
18	AxonFramework-4.11.1/docs/_reference/modules/legacy/pages	AxonFramework-4.11.1/docs/_reference/modules/legacy
19	AxonFramework-4.11.1/docs/_reference/modules/legacy/partials	AxonFramework-4.11.1/docs/_reference/modules/legacy
20	AxonFramework-4.11.1/docs/_reference/modules/metrics/pages	AxonFramework-4.11.1/docs/_reference/modules/metrics
21	AxonFramework-4.11.1/docs/_reference/modules/metrics/partials	AxonFramework-4.11.1/docs/_reference/modules/metrics
22	AxonFramework-4.11.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.11.1/docs/_reference/modules/micrometer
23	AxonFramework-4.11.1/docs/_reference/modules/micrometer/partials	AxonFramework-4.11.1/docs/_reference/modules/micrometer
24	AxonFramework-4.11.1/docs/_reference/modules/modeling/partials	AxonFramework-4.11.1/docs/_reference/modules/modeling
25	AxonFramework-4.11.1/docs/_reference/modules/spring/pages	AxonFramework-4.11.1/docs/_reference/modules/spring
26	AxonFramework-4.11.1/docs/_reference/modules/spring/partials	AxonFramework-4.11.1/docs/_reference/modules/spring
27	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter
28	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter
29	AxonFramework-4.11.1/docs/_reference/modules/test/pages	AxonFramework-4.11.1/docs/_reference/modules/test

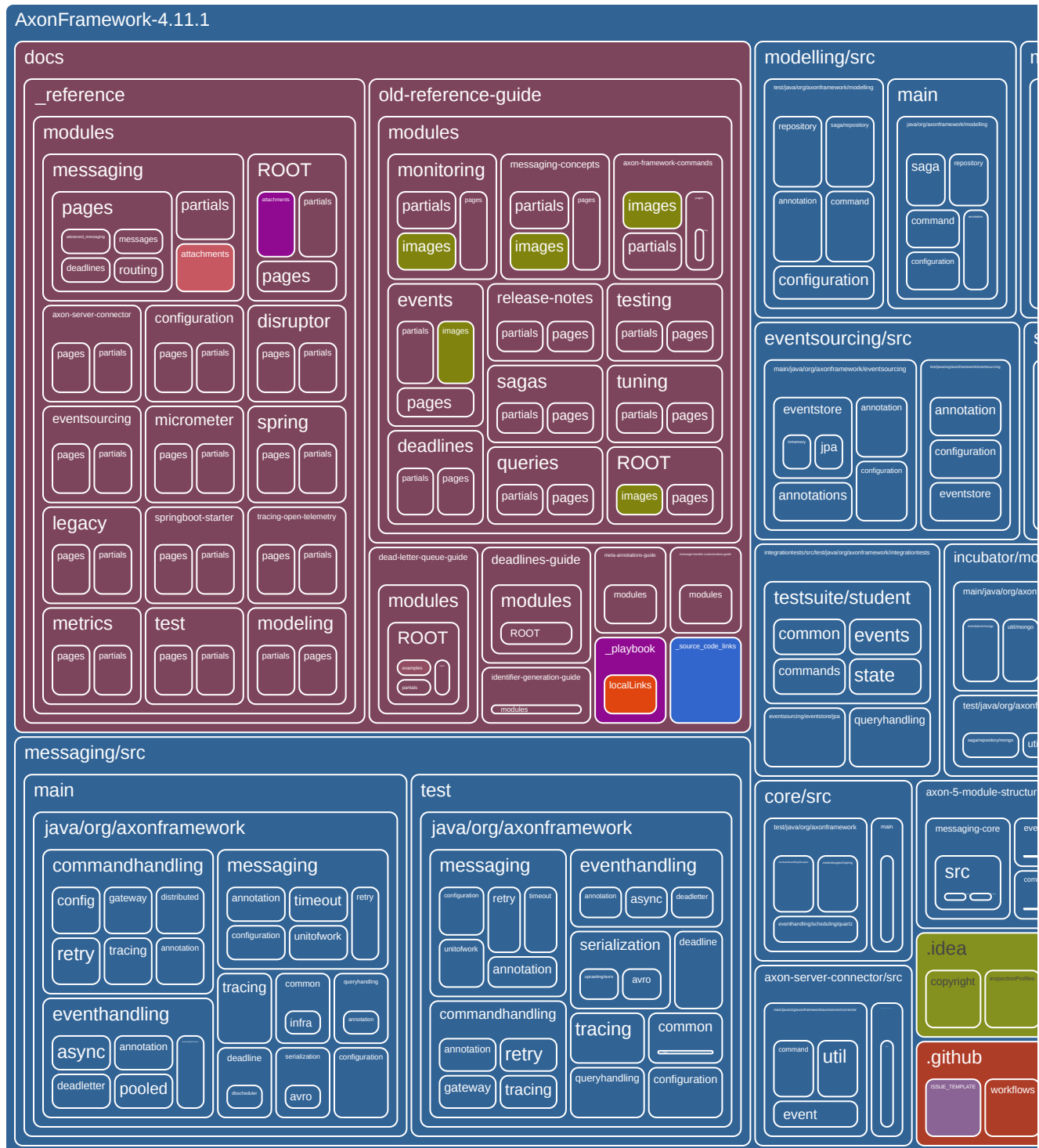
Number of files per directory

Directories and their file count



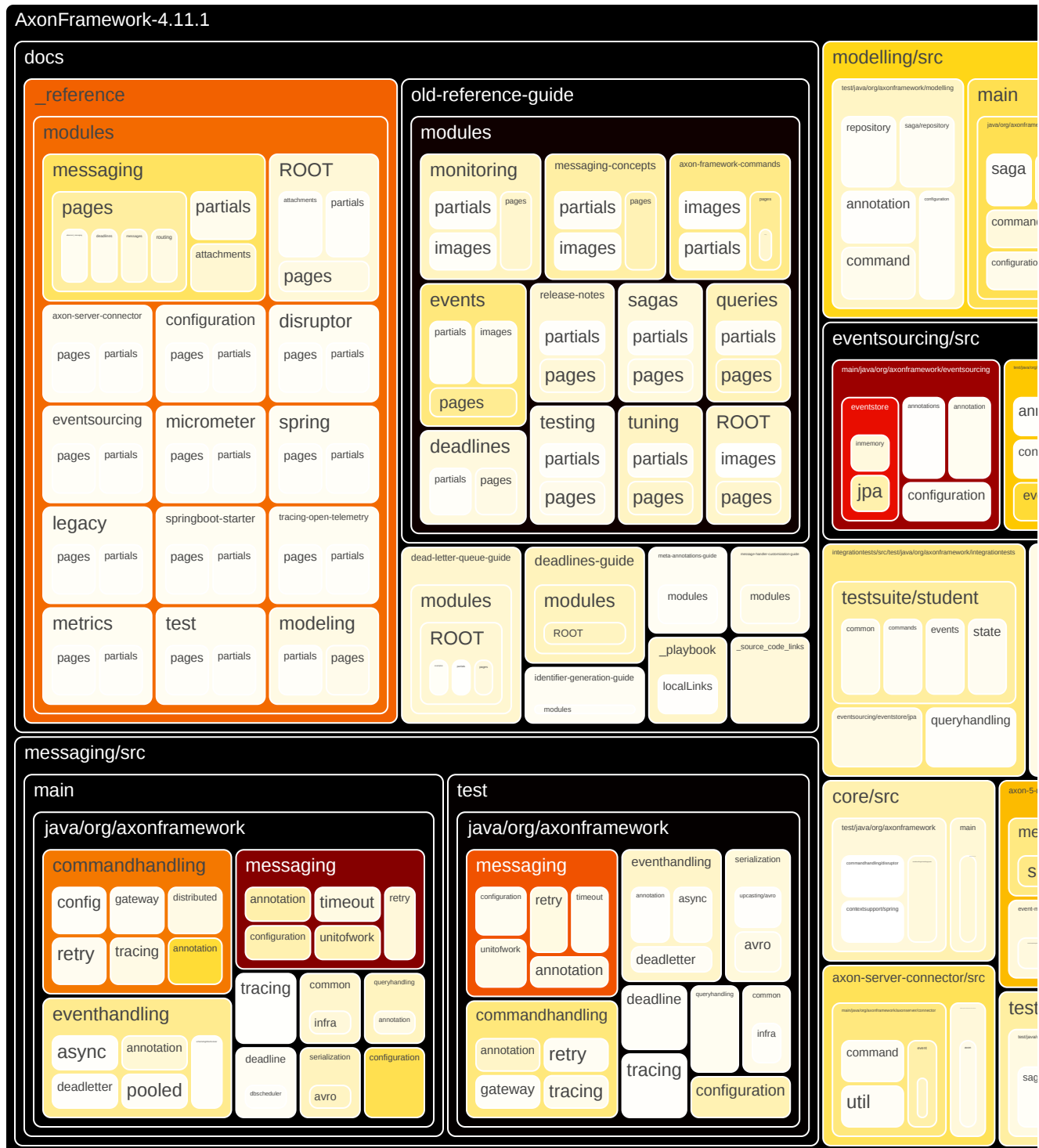
Most frequent file extension per directory

Most frequent file extension per directory



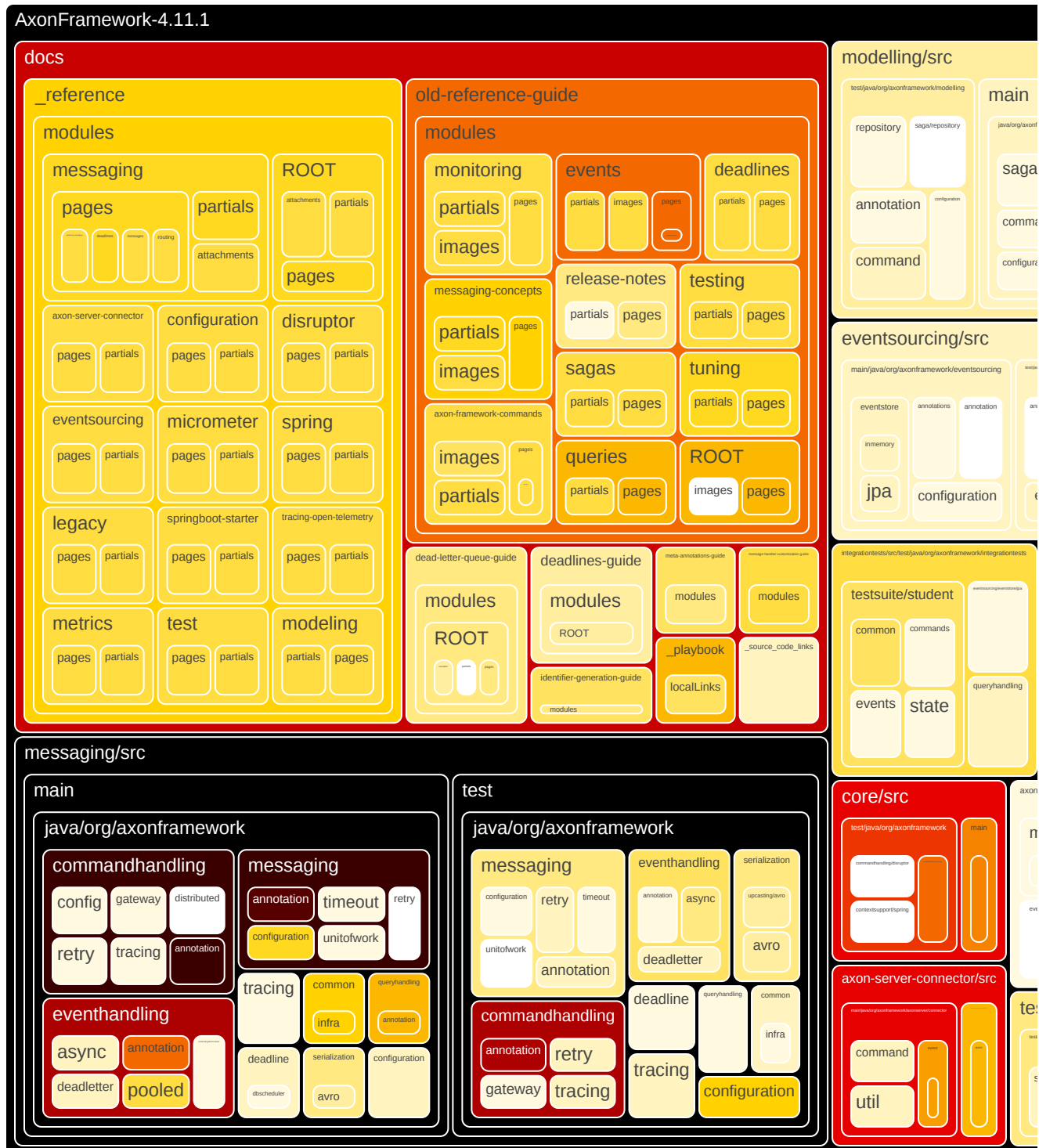
Number of commits per directory

Number of git commits



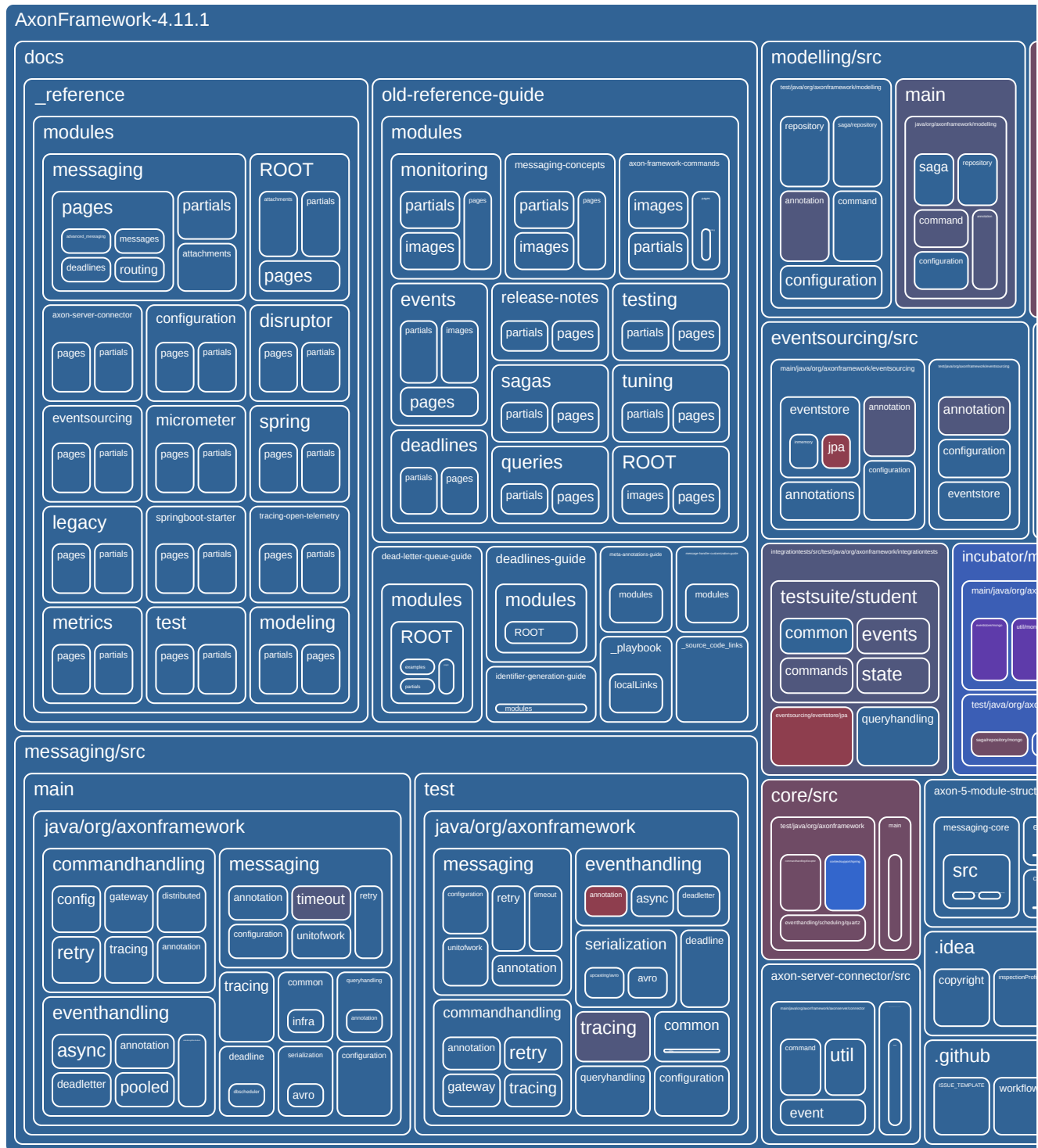
Number of distinct authors per directory

Number of distinct commit authors



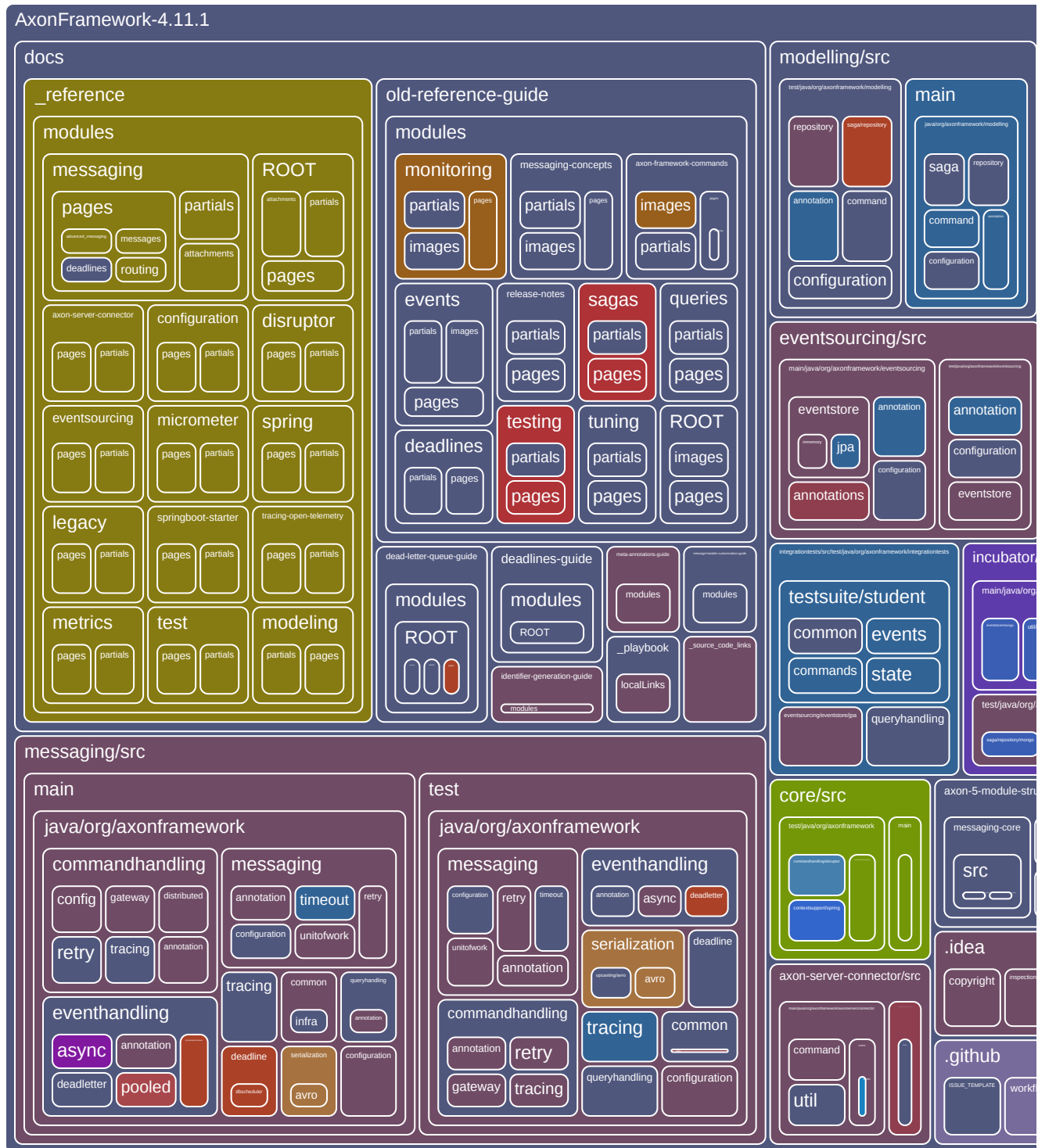
Main author per directory

Main authors with highest number of commits



Second author per directory

Second author with the second highest number of commits



Days since last commit per directory

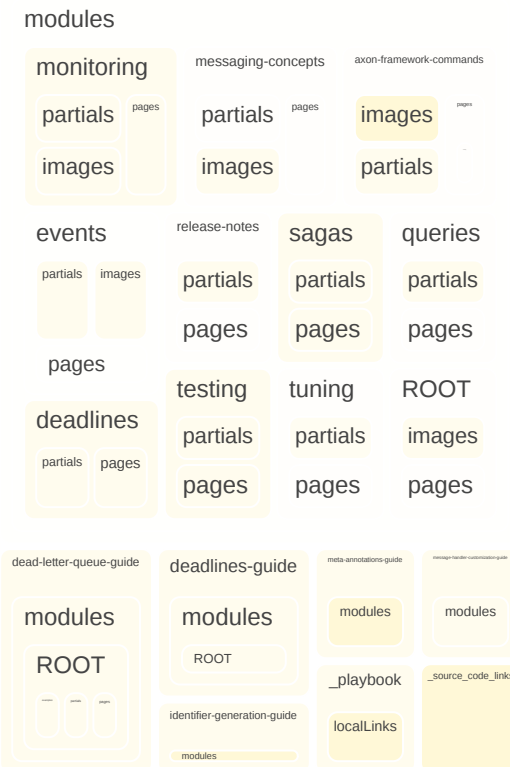
Days since last commit

AxonFramework-4.11.1

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old-reference-guide



modelling/src



eventsourcing/src



integrationtests/src/test/java/org/axonframework/integrationtests

testsuite/student



messaging/src

main

java/org/axonframework

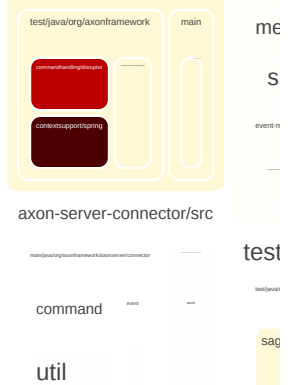


test

java/org/axonframework



core/src



axon-server-connector/src

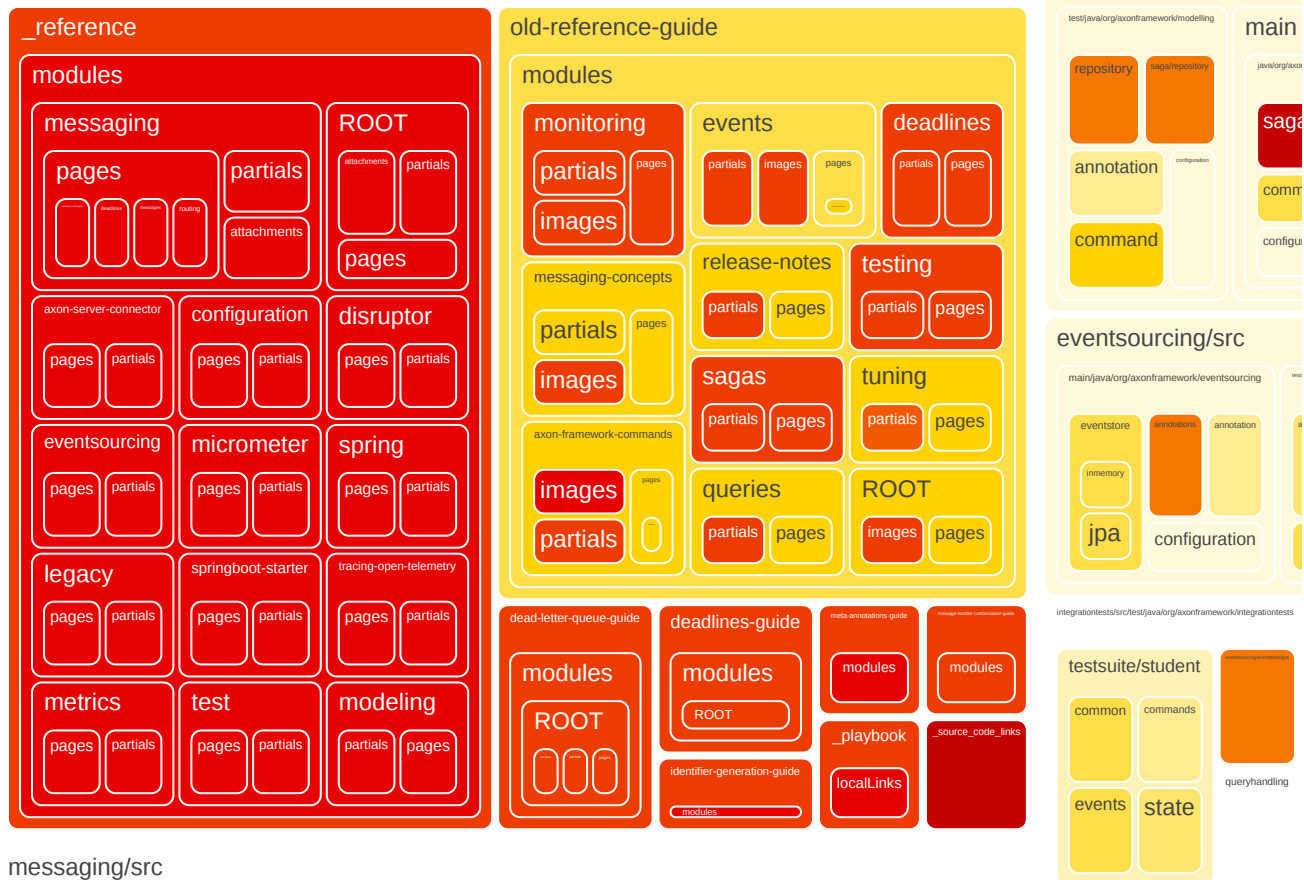


Days since last commit per directory (ranked)

Rank of days since last commit

AxonFramework-4.11.1

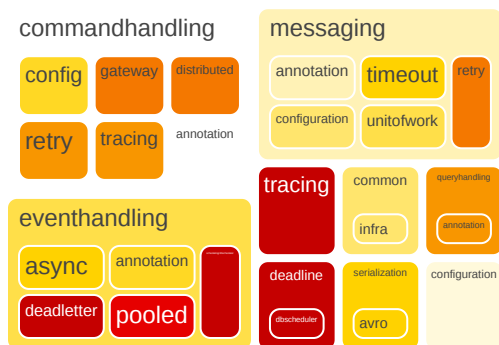
docs



messaging/src

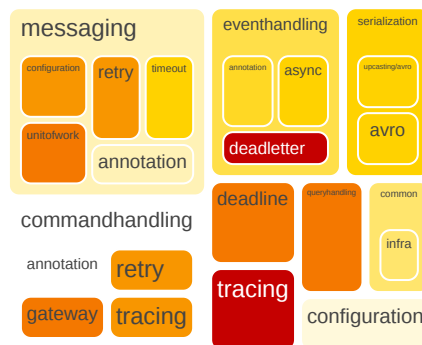
main

java/org/axonframework

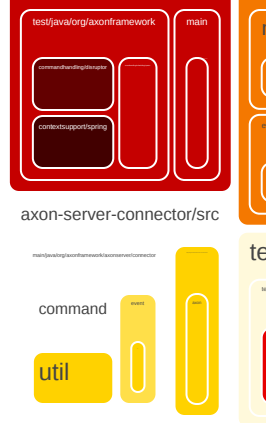


test

java/org/axonframework



core/src



Days since last file creation per directory

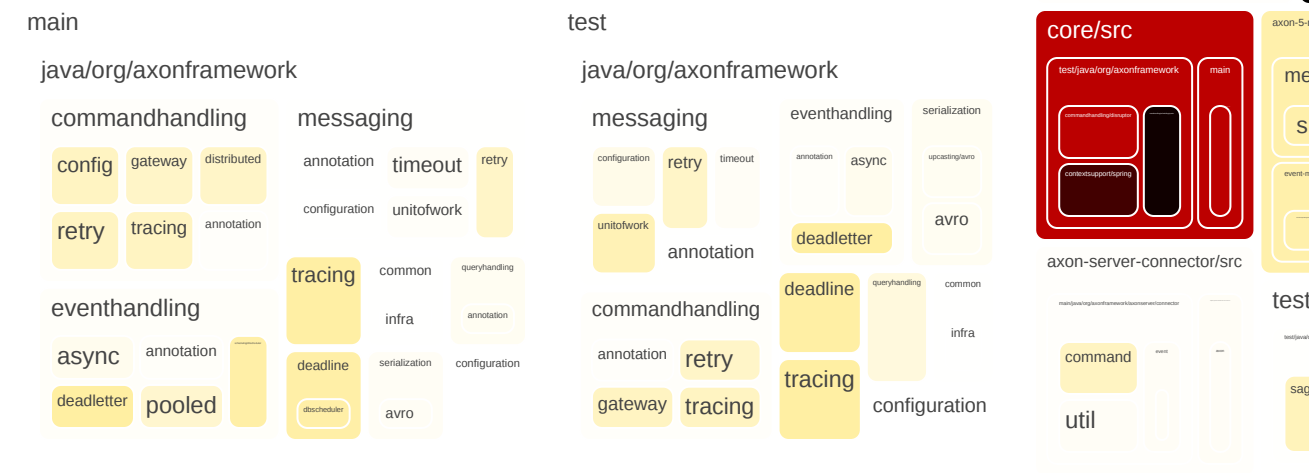
Days since last file creation

AxonFramework-4.11.1

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messaging/src

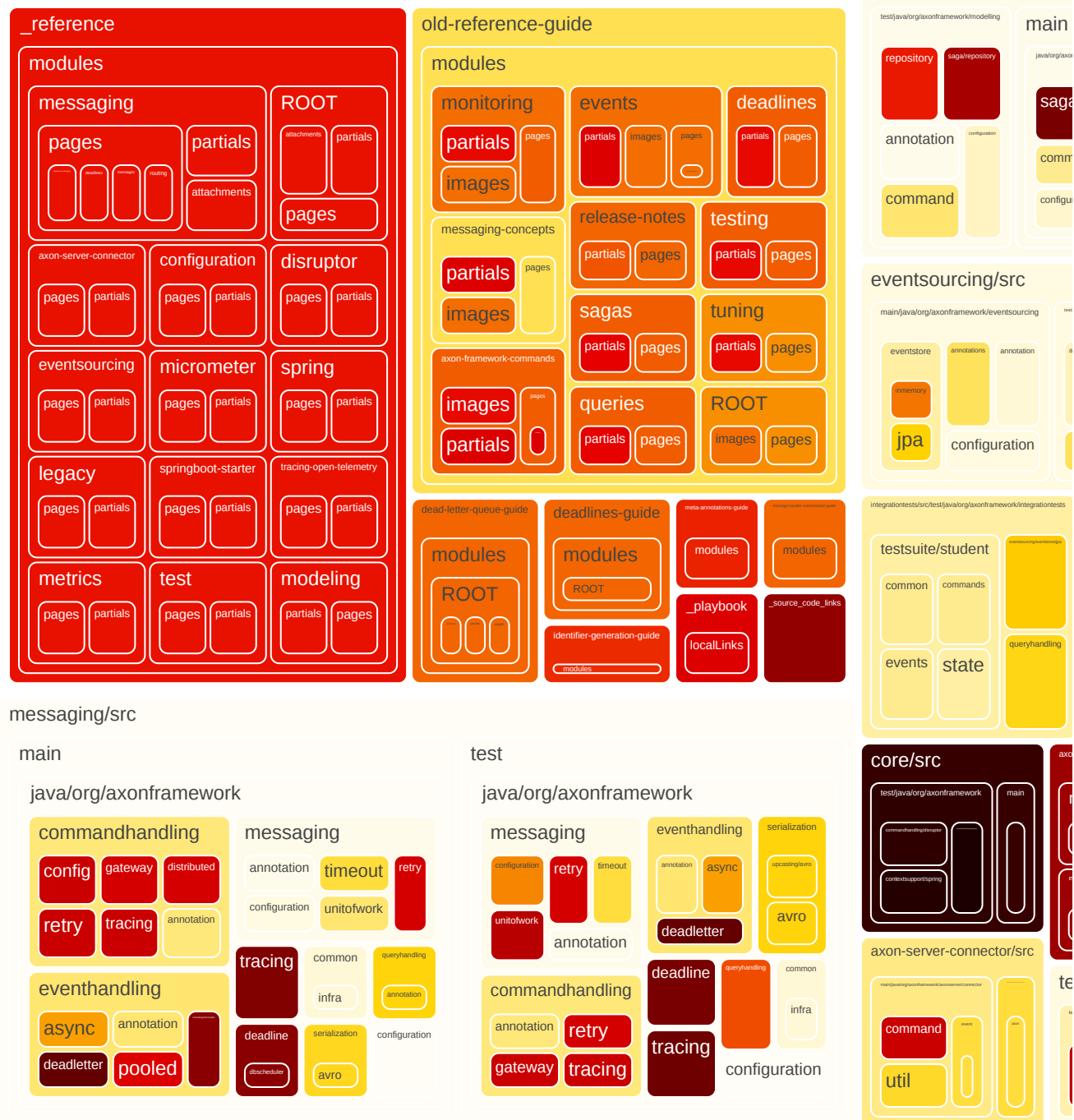


Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.1

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Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.1

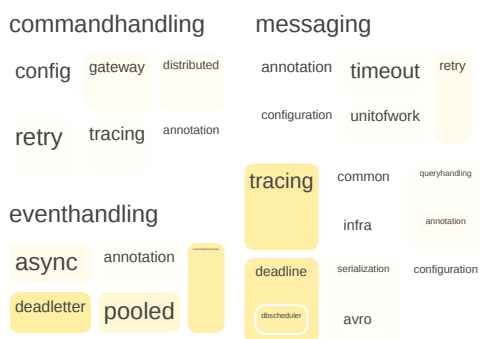
docs



messaging/src

main

java/org/axonframework



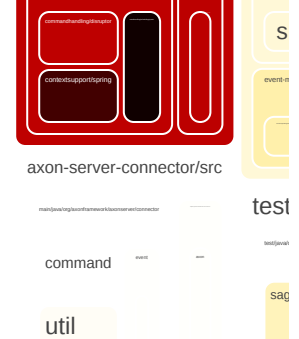
test

java/org/axonframework



core/src

test/java/org/axonframework

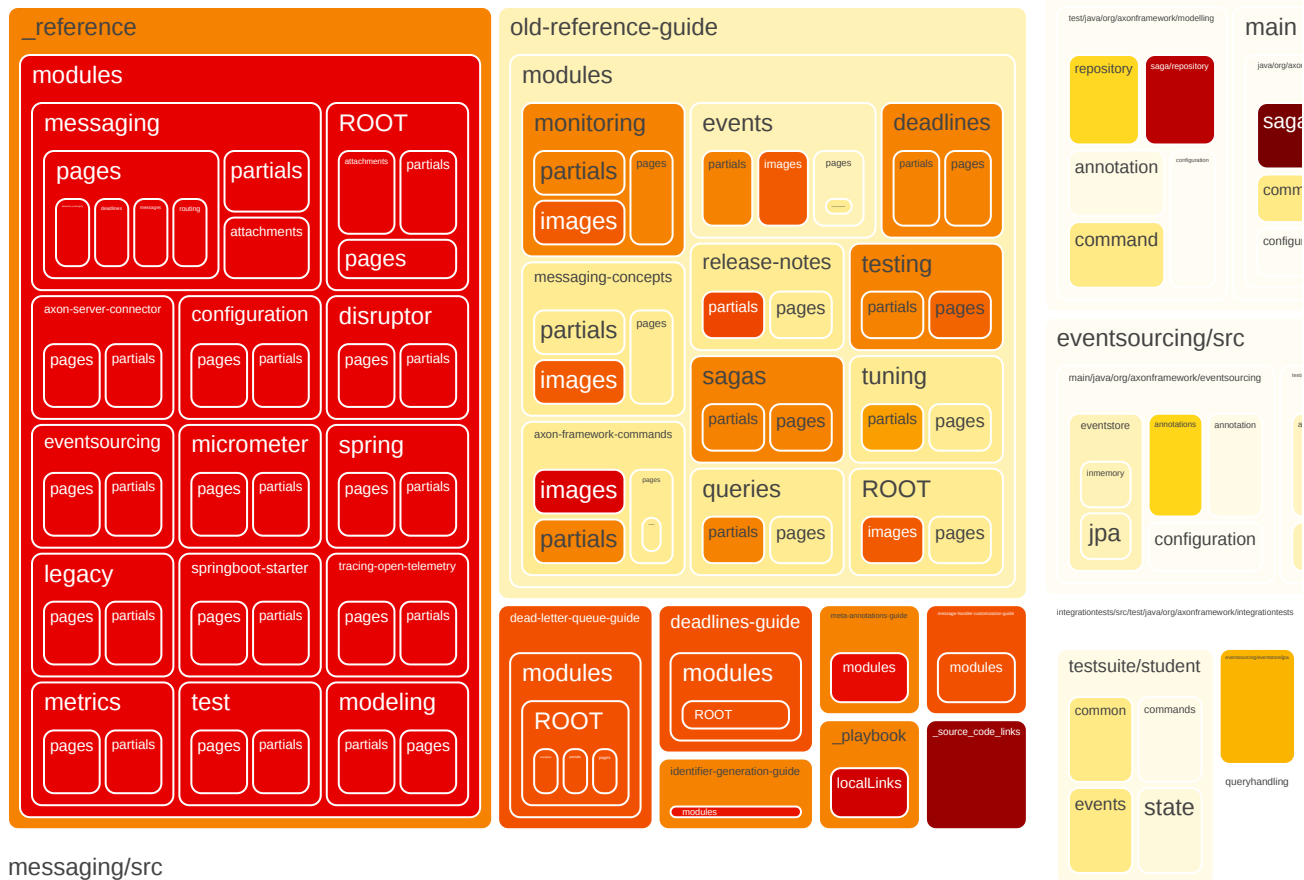


Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-4.11.1

docs



messaging/src

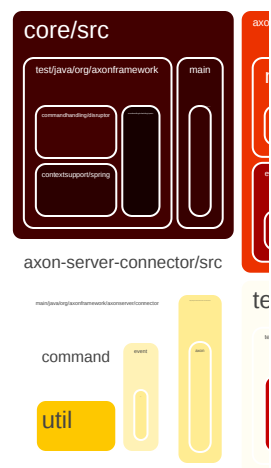
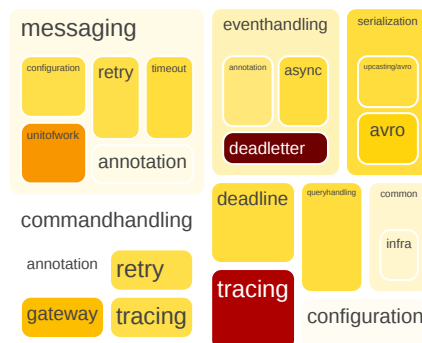
main

java/org/axonframework



test

java/org/axonframework



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

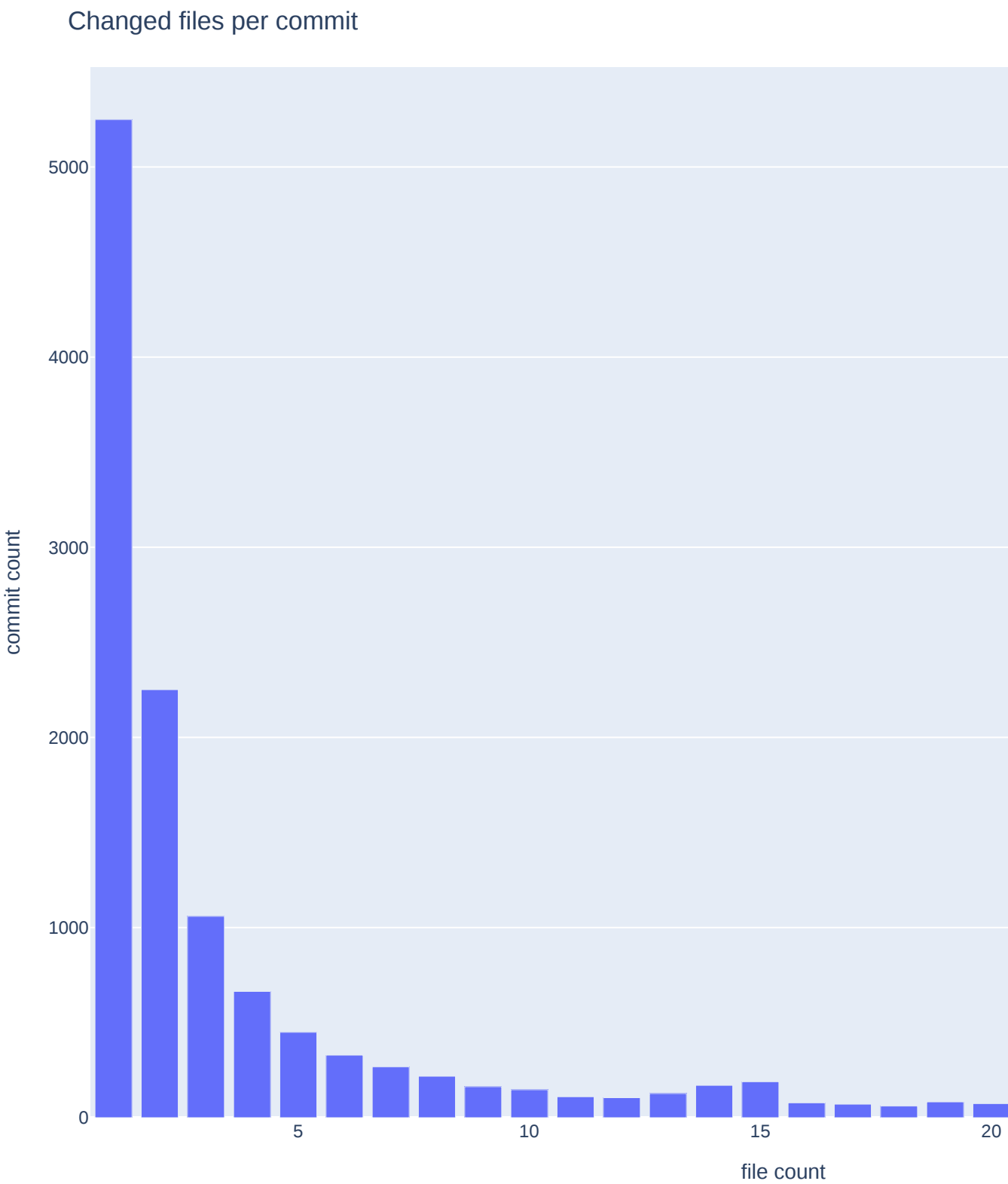
Sum of commits that changed more than 30 files (each) = 1181

Max changed files with one commit = 4041

	filesPerCommit	commitCount
count	377.000000	377.000000
mean	398.572944	35.607427
std	559.546155	302.200493
min	1.000000	1.000000
25%	95.000000	1.000000
50%	204.000000	2.000000
75%	429.000000	5.000000
max	4041.000000	5249.000000

	filesPerCommit	commitCount
0	1	5249
1	2	2251
2	3	1059
3	4	664
4	5	449
5	6	329
6	7	267
7	8	218
8	9	163
9	10	147
10	11	110
11	12	105
12	13	127
13	14	170
14	15	189
15	16	78
16	17	71
17	18	61
18	19	83
19	20	74
20	21	46
21	22	68
22	23	34
23	24	38
24	25	43
25	26	33
26	27	34
27	28	30
28	29	29
29	30	24

Bar chart with the number of files per commit distribution



WordCloud of git authors

	word	frequency
0	Steven van Beelen	4357
1	Allard Buijze	3156
2	smcvb	822
3	Mateusz Nowak	506
4	Milan Savic	449
5	Mitchell Herrijgers	438
6	Gerard Klijs	247
7	Marc Gathier	231
8	sara pellegrini	163
9	Rene de Waele	129

Wordcloud of git authors

